

Section 0 - Preliminary Checks

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Description

This is the first of four analysis files. This file covers the descriptive statistic tables that show demographic differences between Steam-Only and Dual-Platform games in the sample. There are three main tables. The first table compares Steam-Only and Dual-Platform games' average price, real price, sale frequency and discount depth before and after Epic entered the market. The second table compares the release year distribution between the two groups, and the third table compares genre composition.

Notes / Important Information

- In Table 1, Pre-Epic includes all games released before Dec 2018 (entry date). All Dual-Platform games in this section are games that will become Dual-Platform in the future once Epic launches. As a result of this, the sample size for Pre-Epic is smaller than the Post-Epic sample size. Additionally, this affects the total sample values since Pre-Epic contains fewer games than the Post-Epic and Full Sample sections that include all games from the sample.
 - Table 1's averages and t-tests are both computed at the game level (one average per game per period). Every game contributes equally regardless of how many days of price data it has.
 - Table 1 stars come from two-sample t-tests comparing the distribution of game-level means between groups.
 - Tables 2 and 3 stars come from chi-square tests of independence between group and the attribute
 - Tables 2 and 3 describe time-invariant game attributes (release year, genre) and also use one row per game.
-

Setup and Data Cleaning

```
# load libraries
library(tidyverse)
library(stargazer)

# load main dataset
data <- read_csv("Data/cleaned_data.csv")

# establish epic launch date
epic_launch <- as.Date("2018-12-06")

# establish genre columns
```

```
genre_cols <- c("RPG", "Action", "Adventure", "Shooter", "Strategy", "Simulation",
               "Platformer", "Puzzle", "Horror", "OpenWorld", "Survival",
               "Roguelike", "Multiplayer", "Narrative")
```

Epic Entry Dates

First, I have to obtain the date each Dual-Platform game entered Epic since game has its own entry timeline.

```
epic_entry_dates <- data %>%
  filter(shop_name == "Epic") %>%
  group_by(game_title) %>%
  summarise(entry_date = min(date), .groups = "drop")
```

Build Base Steam Panel

Next, I create the base steam dataset while joining the Epic entry dates

```
steam_base <- data %>%
  filter(shop_name == "Steam") %>%
  # Join Epic entry dates
  left_join(epic_entry_dates, by = "game_title") %>%
  mutate(
    # Variable to track if game has ever been on Epic
    ever_on_epic = !is.na(entry_date),
    # Variable that tracks Steam-only vs Dual-Platform
    group = if_else(ever_on_epic, "Dual-Platform", "Steam-Only"),
    # Variable for release year
    release_year = year(release_date)
  )
```

Game-Level Summaries — one row per game per period

These are used for both the displayed averages and the t-tests in Table 1.

```
# Helper to collapse steam_base to game level for a given date filter
make_game_summary <- function(df) {
  df %>%
    group_by(game_title, group, release_year) %>%
    summarise(
      avg_price = mean(price, na.rm = TRUE),
      avg_real_price = mean(real_price, na.rm = TRUE),
      avg_sale_freq = mean(sale, na.rm = TRUE),
      avg_disc_depth = mean(discount_percent, na.rm = TRUE),
      across(all_of(genre_cols), first),
      .groups = "drop"
    )
}

# Pre-Epic: only observations before Dec 2018
game_summary_preepic <- steam_base %>%
  filter(date < epic_launch) %>%
  make_game_summary()

# Post-Epic: only observations after Dec 2018
game_summary_postepic <- steam_base %>%
```

```

filter(date >= epic_launch) %>%
make_game_summary()

# Full Sample: all observations
game_summary <- steam_base %>%
  make_game_summary()

# Game counts for each time group
n_so_pre <- sum(game_summary_preepic$group == "Steam-Only")
n_dual_pre <- sum(game_summary_preepic$group == "Dual-Platform")
n_so_post <- sum(game_summary_postepic$group == "Steam-Only")
n_dual_post <- sum(game_summary_postepic$group == "Dual-Platform")
n_so_full <- sum(game_summary$group == "Steam-Only")
n_dual_full <- sum(game_summary$group == "Dual-Platform")

```

Table 1: Outcome Variable Comparison

Game-level averages displayed and tested. Each game contributes one mean per period. Two-sample t-test compares Dual-Platform vs Steam-Only distributions of those game means.

```

# Row builder that takes the game-level summary for one period
make_row <- function(var, label, game_df) {
  # Pull out the variable vectors for each group
  so_vec <- game_df[[var]][game_df$group == "Steam-Only"]
  dual_vec <- game_df[[var]][game_df$group == "Dual-Platform"]

  # Compute means for each group
  so_mean <- mean(so_vec, na.rm = TRUE)
  dual_mean <- mean(dual_vec, na.rm = TRUE)

  # Run t-test comparing the two groups
  tt <- t.test(dual_vec, so_vec)

  # Add significance stars based on p-value
  stars <- case_when(
    tt$p.value < 0.01 ~ "***",
    tt$p.value < 0.05 ~ "**",
    tt$p.value < 0.10 ~ "*",
    TRUE ~ ""
  )

  # Return the row as a character vector
  c(label,
    format(round(so_mean, 2), nsmall = 2),
    format(round(dual_mean, 2), nsmall = 2),
    paste0(round(dual_mean - so_mean, 2), stars))
}

# Build combined three-panel matrix
combined_mat <- rbind(
  # Pre-Epic
  c("Pre-Epic Period (before Dec 2018)", "", "", ""),

```

```

make_row("avg_price",      " Avg Price (USD)",      game_summary_preepic),
make_row("avg_real_price", " Avg Real Price (USD)", game_summary_preepic),
make_row("avg_sale_freq",  " Sale Frequency",      game_summary_preepic),
make_row("avg_disc_depth", " Discount Depth (%)", game_summary_preepic),
c(" N (games)", as.character(n_so_pre), as.character(n_dual_pre), ""),
c("", "", "", ""),
# Post-Epic
c("Post-Epic Period (after Dec 2018)", "", "", ""),
make_row("avg_price",      " Avg Price (USD)",      game_summary_postepic),
make_row("avg_real_price", " Avg Real Price (USD)", game_summary_postepic),
make_row("avg_sale_freq",  " Sale Frequency",      game_summary_postepic),
make_row("avg_disc_depth", " Discount Depth (%)", game_summary_postepic),
c(" N (games)", as.character(n_so_post), as.character(n_dual_post), ""),
c("", "", "", ""),
# Full Sample
c("Full Sample Period", "", "", ""),
make_row("avg_price",      " Avg Price (USD)",      game_summary),
make_row("avg_real_price", " Avg Real Price (USD)", game_summary),
make_row("avg_sale_freq",  " Sale Frequency",      game_summary),
make_row("avg_disc_depth", " Discount Depth (%)", game_summary),
c(" N (games)", as.character(n_so_full), as.character(n_dual_full), "")
)

# Set column names for the combined matrix
colnames(combined_mat) <- c("Variable", "Steam-Only",
                           "Dual-Platform", "Difference")

# Render the combined matrix as a LaTeX table using stargazer
stargazer(combined_mat,
           type      = "latex",
           title     = "Outcome Variable Comparison by Group",
           summary   = FALSE,
           rownames  = FALSE,
           notes     = "**** p<0.01, ** p<0.05, * p<0.10",
           notes.align = "l")

```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac@gmail.com % Date and time: Tue, Sep 22, 2026 - 23:19:32

Table 2: Release Year Distribution

Time-invariant attribute. Uses full-sample game_summary so every game appears once. Chi-square test per year for compositional differences.

```

# Gather all unique release years in the dataset
all_years <- sort(unique(game_summary$release_year))

# Row builder for release year distribution table
make_year_row <- function(yr) {
  # Compute the percentage of games released in the given year for each group
  so_pct <- mean(game_summary$release_year[game_summary$group == "Steam-Only"] == yr,
                 na.rm = TRUE) * 100
}

```

Table 1: Outcome Variable Comparison by Group

Variable	Steam-Only	Dual-Platform	Difference
Pre-Epic Period (before Dec 2018)			
Avg Price (USD)	23.74	28.21	4.47***
Avg Real Price (USD)	22.88	27.19	4.31***
Sale Frequency	0.12	0.14	0.02**
Discount Depth (%)	5.54	5.62	0.08
N (games)	335	174	
Post-Epic Period (after Dec 2018)			
Avg Price (USD)	24.03	27.67	3.64***
Avg Real Price (USD)	19.96	22.63	2.67***
Sale Frequency	0.19	0.22	0.03***
Discount Depth (%)	10.32	11.92	1.6***
N (games)	544	356	
Full Sample Period			
Avg Price (USD)	24.56	28.39	3.83***
Avg Real Price (USD)	20.87	23.68	2.81***
Sale Frequency	0.18	0.21	0.03***
Discount Depth (%)	9.28	10.85	1.57***
N (games)	544	356	

*** p<0.01, ** p<0.05, * p<0.10

```
dual_pct <- mean(game_summary$release_year[game_summary$group == "Dual-Platform"] == yr,
                 na.rm = TRUE) * 100

# Run chi-square test for the given year
tbl <- table(game_summary$group, game_summary$release_year == yr)
chi <- suppressWarnings(chisq.test(tbl))

# Add significance stars based on p-value
stars <- case_when(
  chi$p.value < 0.01 ~ "***",
  chi$p.value < 0.05 ~ "**",
  chi$p.value < 0.10 ~ "*",
  TRUE ~ ""
)

# Return the row as a character vector
c(as.character(yr),
  paste0(round(so_pct, 1), "%"),
  paste0(round(dual_pct, 1), "%"),
  paste0(round(dual_pct - so_pct, 1), stars))
}

# Build the release year distribution table
table2_mat <- do.call(rbind, lapply(all_years, make_year_row))
colnames(table2_mat) <- c("Release Year", "Steam-Only (%)",
                        "Dual-Platform (%)", "Difference")
```

```
# Render the release year distribution table as a LaTeX table using stargazer
stargazer(table2_mat,
  type      = "latex",
  title      = "Release Year Distribution by Group",
  summary    = FALSE,
  rownames   = FALSE,
  notes      = "*** p<0.01, ** p<0.05, * p<0.10",
  notes.align = "l")
```

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Table 2: Release Year Distribution by Group

Release Year	Steam-Only (%)	Dual-Platform (%)	Difference
2013	0.6%	0%	-0.6
2014	11.4%	6.5%	-4.9**
2015	12.5%	8.4%	-4.1*
2016	11.6%	9.6%	-2
2017	10.7%	12.1%	1.4
2018	11.4%	10.4%	-1
2019	9.2%	9%	-0.2
2020	6.6%	9.8%	3.2
2021	7.7%	11.8%	4.1*
2022	9%	10.7%	1.7
2023	9.4%	11.8%	2.4

*** p<0.01, ** p<0.05, * p<0.10

Table 3: Genre Distribution

Time-invariant attribute. Uses full-sample game_summary so every game appears once. Genres sorted by absolute difference between groups.

```
# Row builder for genre distribution table
make_genre_row <- function(g) {
  # Compute the percentage of games in the given genre for each group
  so_pct <- mean(game_summary[[g]][game_summary$group == "Steam-Only"],
    na.rm = TRUE) * 100
  dual_pct <- mean(game_summary[[g]][game_summary$group == "Dual-Platform"],
    na.rm = TRUE) * 100

  # Run chi-square test for the given genre
  tbl <- table(game_summary$group, game_summary[[g]])
  chi <- suppressWarnings(chisq.test(tbl))

  # Add significance stars based on p-value
  stars <- case_when(
    chi$p.value < 0.01 ~ "***",
    chi$p.value < 0.05 ~ "**",
    chi$p.value < 0.10 ~ "*",
    TRUE ~ "")
}
```

```

    TRUE ~ ""
  )

  # Return the row as a list containing genre, difference, and row data
  list(
    genre = g,
    diff = dual_pct - so_pct,
    row = c(g,
             paste0(round(so_pct, 1), "%"),
             paste0(round(dual_pct, 1), "%"),
             paste0(round(dual_pct - so_pct, 1), stars))
  )
}

# Build the genre distribution table, sorting by absolute difference
genre_rows <- lapply(genre_cols, make_genre_row)
genre_rows_sorted <- genre_rows[order(sapply(genre_rows, function(x) abs(x$diff)),
                                         decreasing = TRUE)]

# Combine the sorted rows into a matrix for stargazer
table3_mat <- do.call(rbind, lapply(genre_rows_sorted, function(x) x$row))
colnames(table3_mat) <- c("Genre", "Steam-Only (%)",
                          "Dual-Platform (%)", "Difference")

# Render the genre distribution table as a LaTeX table using stargazer
stargazer(table3_mat,
           type = "latex",
           title = "Genre Distribution by Group",
           summary = FALSE,
           rownames = FALSE,
           notes = "*** p<0.01, ** p<0.05, * p<0.10",
           notes.align = "l")

```

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Table 3: Genre Distribution by Group

Genre	Steam-Only (%)	Dual-Platform (%)	Difference
Adventure	21.3%	31.2%	9.9***
OpenWorld	16.9%	21.9%	5*
RPG	30.5%	25.6%	-5
Narrative	15.4%	20.2%	4.8*
Multiplayer	20.2%	17.7%	-2.5
Survival	11.2%	13.5%	2.3
Platformer	9.6%	11.5%	2
Shooter	11%	9.3%	-1.8
Horror	8.1%	9.8%	1.7
Strategy	12.9%	14.6%	1.7
Puzzle	5.3%	6.7%	1.4
Action	32.2%	30.9%	-1.3
Simulation	13.1%	13.5%	0.4
Roguelike	7.2%	7%	-0.1

*** p<0.01, ** p<0.05, * p<0.10